An overview of qualitative research in open science: literature review

Alejandra Manco
Université Claude Bernard Lyon 1

Transcending Boundaries in Qualitative Research
September 22-23, 2021

Introduction

The concept of open science is gaining traction around the world.

Purpose

To provide an overview of the essential characteristics of the qualitative empirical articles on open science that have been published to date.

Methodology

The searches were made in Google Scholar, Semantic Scholar and Scopus.

Keyword: open science.

Texts in English, Spanish, Portuguese and French.

Scopus

A total of 28 articles were selected.

Results

1) Which is the most used technique for researching in open science?

Interviews

Researcher's opinion or perspective about some specific aspect of open science

Participants' encounters, opinions, and knowledge gaps.

Document nalysis

Open science policies:

Journal policies and institutional or official governmental policies on various facets of open science

2) Which type of research design did the study used?

- Descriptive
- Exploratory

3) Which component of open science was the focus of the research?

Open science policies and science policy.

Other works tend to focus on open data and open science practice and how open science is intertwined in these issues.

4) Which are the areas that have developed the most qualitative research on open science?

The sciences and technology are the topics in which the majority empirical open science works have been performed.

5) Which sampling strategy was used?

Lack of a formal sampling methodology.

Often only very seldom papers mention the specific sampling criteria for choosing their research subjects or objects.

When mentioned, the most common sampling strategies are *snowball, purposive, criterion, convenience and self-selective* sampling and their distinctive combinations.

6) Which is the sample used in the study?

Use of a small interview sample and some entities (laboratories, foundations, or initiatives) when working with case study, in most of the cases.

The content corpus of documents – such as policies or guidelines – is typically larger than the content corpus of interviews, but it may be further composed of two main groups: a relatively small corpus -around 100 documents- and a corpus of thousands of documents.

7) Which was the unit of analysis of the studies?

A common feature is the fact that other researchers are the unit of analysis when working with interviews.

Document analysis or content analysis analytical approaches are commonly seen in qualitative studies on open science that use documents as the unit of analysis.

The last group analyzes entities such as laboratories, foundations, or initiatives. The fact that the number of entities is limited is a common feature.

8) What are the study limitations?

Related to the specific type of research design and the nature of qualitative research.

Exploratory research has issues on the representativeness of the sample and the comprehensiveness of the research or rather the lack of exhaustivity.

Issues in terms of generalization are also brought up: related to the chosen language, countries, studied areas or the lack of representativeness of the sampling criteria instrument.

9) Is research data available for these studies?

The vast majority of the qualitative empirical corpus reviewed does not publish their research data.

The few studies that have research data available usually have it published on international platforms such as Zenodo or Figshare.